

OPENGUT: An Open-Source Platform for Gut-Computer Interaction Research

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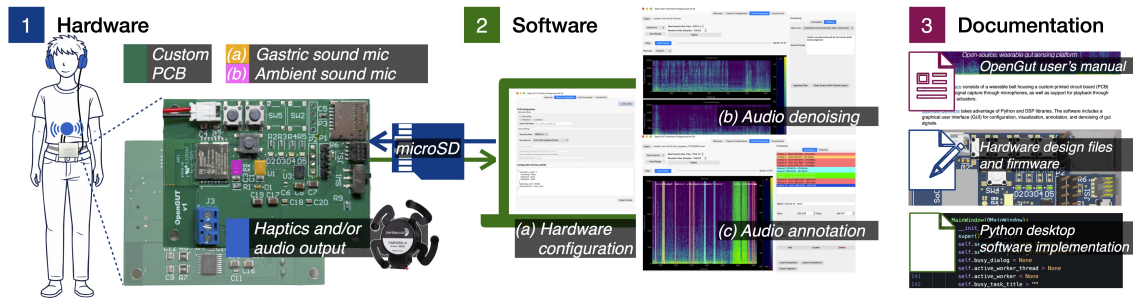


Figure 1: Overview of the OPENGUT platform: hardware (custom PCB with audio input, and audio/haptic output), software GUI (configuration setup, data analysis and annotation), and extensive documentation (user manual, design files and codebase).

Abstract

Researchers are increasingly exploring the gut as a human-computer interface due to its gut-brain axis, linking physiological states with emotion and cognition. Despite its potential, the gut research space remains highly fragmented and underexplored. This is largely due to the absence of open and accessible hardware and software tools that could enable systematic investigation across disciplines and

communities. To address this gap, we conducted semi-structured interviews with six experts to identify pain points in using existing platforms. Based on the themes identified, we formulated a set of requirements, and developed OPENGUT, a rapid prototyping platform for gut research. Our open source platform consists of three main components: a wearable hardware platform for data capture and playback, a graphical user interface for visualization and analysis, as well as [accompanying documentation for platform setup and customization](#). We perform technical evaluation of OPENGUT for its signal quality, noise suppression, as well as [outline potential usage scenarios for gut research](#).

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CCS Concepts

• Human-centered computing → Systems and tools for interaction design.

Keywords

Gut-computer interaction, open-source platform, bowel sounds

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1 Introduction

The gastrointestinal system, or the *gut*, is central to both human physiology and psychology, as it is responsible for digestion and hosts the *enteric nervous system*, which is the second largest nervous system in the human body [8, 14, 15, 35]. This uniqueness has piqued the interest of HCI researchers, with several gut-related research projects emerging in recent years. **Broadly, prior work has pursued two directions: using gut signals to characterize psychological states [6, 27, 41, 44] as well as providing them as feedback to understand how people perceive internal body signals [26, 31, 33, 36].**

While the possibilities seem endless, the gut research space remains fragmented, with tools confined to isolated use cases and lacking integration [20, 25]. This has resulted in an absence of shared understanding amongst researchers and has heightened the entry barrier for new researchers. In particular, existing gut platforms provide several capabilities for recording bowel sounds and electrogastrography (EGG) signals, but are application-specific and often lack documentation for replication [5, 45, 47].

To better understand the challenges faced by gut researchers, we conducted a formative study with six researchers through semi-structured interviews. Through these, we identified four themes that highlight the challenges in existing platforms, including wearability issues, as well as insufficient hardware and software features for research purposes. Based on the gaps identified from the literature and our formative study, we developed OPENGUT, an open-source platform¹ for gut-computer interaction research. OPENGUT consists of three pillars, as shown in Figure 1. OPENGUT hardware consists of a wearable belt that houses a custom printed circuit board (PCB), enabling signal capture through microphones and playback through haptic and audio actuators. OPENGUT software includes a graphical user interface (GUI) for configuration, visualization, annotation and denoising of gut signals. **We also release hardware designs, firmware, software and documentation as resources for future replication and community-driven development.** In addition, OPENGUT supports multiple modes of operation, including logging, playback, and real-time loopback, enabling a range of experimental and interactive scenarios.

We evaluated OPENGUT to demonstrate its signal quality, as well as its long-term recording capabilities. Specifically, we compared the gut sounds recorded by OPENGUT to a medical-grade stethoscope, and obtained high correlation scores for over 75% of the audio segments containing gut activity, demonstrating its efficacy. We also illustrate OPENGUT's ability to capture variations in gut activity in longitudinal settings, as well as the ability to support 24 hours of

recording with a small battery capacity of 1000 mAh, showcasing its suitability for real-world settings.

Overall, this paper contributes OPENGUT, an enabling tool for the emerging gut-computer interaction space, where hardware and software are currently fragmented.

2 Related Work

Gut computer interaction is emerging as a novel research space within the HCI community, with initial efforts focusing on understanding mental well-being [6, 27, 41, 44], as well as manipulating or understanding interoception (i.e., a user's ability to perceive internal bodily states) [26, 31, 33, 36]. With regards to mental well-being, prior work has linked gut activity to affective states [41, 44], demonstrated stress inference from bowel sounds [6], and explored the relationship between bowel sounds and emotion [27]. With regards to interoception, researchers have investigated the effects of rumbling sound playback on food desire [36] and used haptic and auditory feedback to influence gastric interoception, bodily awareness, and self-reflection [26, 31].

For sensing bowel sounds, there are several medical-grade stethoscopes as well as research platforms. Digital stethoscopes [21, 32, 39] provide high-quality audio pickup, noise cancellation, audio playback through headphones, as well as real-time visualization via companion mobile application, making them reliable tools for clinical use. However, they are closed-source, and provide limited support for low-level data access and custom sensing configurations, making it less convenient for research purposes. Although gut researchers have adapted them in the past [26, 27], these stethoscopes are not designed for long-duration, body-worn settings. To overcome these limitations, researchers have proposed several wearable bowel-sound recording solutions each with different form-factors including skin patches [25, 45, 47], smart T-shirt [5] as well as belt [20]. While these platforms extend the hardware and software scope for researchers with regards to support for multiple microphones, longer battery lives and wearability, **not all hardware and software resources required for replication are publicly available**, making it challenging for new researchers interested in gut research. Table 1 compares existing platforms, with regards to features supported, their specifications as well as their reproducibility.

Beyond traditional electrogastrography (EGG) systems [18], there are a few research platforms that focus on reliable signal acquisition [16], wearability [17, 42, 43], as well as on developing low-cost and open-source platforms [3]. Notably, Vujic et al. [43] developed a wearable belt integrating disposable electrodes, amplification, and real-time streaming for continuous gut sensing. Despite these efforts, there is still a need for an open-source platform, with extensive hardware and software support.

3 Formative Study

To better understand current practices and challenges in gut research, we conducted semi-structured interviews with six experts. **Expert Participants.** We contacted 14 experts (11 female) who are the authors of prominent papers in this space. We received responses from six of them (5 female), and interviewed all of them. Their backgrounds included neuroscience (n=2), psychology (n=1), biomedical engineering (n=1), and game design (n=2). They worked

¹<https://opengut.github.io/>

Platform (* = Stethoscope)	Playback (Audio/Haptic)	Form Factor	Open Hardware	Open Software	Documentation	Graphic User Interface
Yoo et al. [47]	–	Skin patch	No	No	Yes	Mobile app
Mansour et al. [25]	–	Skin patch	No	Partial [24]	Yes	–
Li et al. [20]	–	Belt	Partial [19]	Partial	Yes	Mobile app
Baronetto et al. [5]	–	Smart T-shirt	No	No	Yes	–
Wang et al. [45]	–	Skin patch	No	No	Yes	Mobile app
Stemoscope PRO* [32]	Audio	Handheld	No	No	No	DrStemo app
Littmann CORE* [21]	Audio	Handheld	No	No	No	Eko app
Thinklabs* [39]	Audio	Handheld	No	No	No	Thinklabs Wave app
OPENGUT	Audio & Haptic	Belt	Yes	Yes	Yes	Desktop app

Table 1: Comparison of existing gut platforms with regards to features, form-factor as well as open source resources.

with different sensors, including microphones (n=3), EGG electrodes (n=2) and ingestible sensors (n=1), as well as actuators such as heatpads (n=1) and audio playback (n=1). Their research focus spanned key areas such as interoception (n=3), affect prediction (n=1), gut health monitoring (n=1) and novel game design (n=1). They were not compensated for their participation.

Semi-Structured Interviews. The interviews lasted for 45-60 minutes over Zoom. Questions focused on (i) participants’ prior projects, (ii) challenges encountered, (iii) hardware/software tools they used, (iv) replicability of custom building, and (v) gut applications they found compelling. We transcribed all interviews using Whisper large-v3 model [46] and manually corrected for errors.

Interview-guided Themes. We identified four recurring themes in the interview data [7], presented below, which, together with the literature, informed the design requirements in Appendix A.

Theme 1: Lack of Wearable Platform. Five out of six experts expressed challenges with the long-term wearability of existing devices, citing large form-factor, poor sensor contact and battery limitations. E6 shared their ideal EGG device’s form-factor – ‘device that is ... wide and flat, and has a low profile so that you can easily wear it underneath clothing’. E2 mentioned challenges with keeping microphone in contact with the skin – ‘[it is hard to] make sure that when you place it [microphone] on the abdomen, they kind of stay’. E4 also mentioned issues with limited battery of a digital stethoscope they used (Stemoscope [32]) – ‘[for] long-term study, longitudinal study, you will need definitely a bigger battery’ and ‘[battery life of] one and a half hours just gets over in like no time’.

Theme 2: Lack of Software Support. Four out of six experts mentioned useful software features that are currently missing such as noise suppression, visualization and real-time sensing. With regards to ambient noise, E2 mentioned factors such as the microphone’s ability to pick up long-range sounds and the presence of external noise in recordings – ‘microphones are ... for far field applications’ and ‘you would hear people ... talking in another room’. E5 mentioned a feature they would have liked for interoception, in addition to audio playback – ‘[audio signal] visually presented almost like Spotify ... like a a moving window of the sound being presented on the screen’. E6 also spoke about being able to visualize frequency composition in real-time for EGG signals – ‘if the software can just, every five to ten seconds... tell you the dominant frequency band at that time’.

Theme 3: Lack of Research-Specific Platform. Three out of six researchers developed their own custom platforms due to the lack of existing solutions that met their requirements. In particular, E6 mentioned the lack of publicly available EGG devices – ‘we don’t have one reliable, publicly available EGG device. So what do you do instead? DIY [do-it-yourself] electronics’. E1 also highlighted the deficiencies of existing platforms for their research needs – ‘[ingestible sensors are] originally for medical purpose’ and ‘[as] HCI researchers, we have different needs’.

Theme 4: Choice of Capturing Bowel Sounds vs. EGG Signals. Researchers recorded gut signals non-invasively through bowel sounds (n=3), and EGG signals (n=2). Four out of six experts mentioned their choice of sensing modality to be influenced by prior works or other researchers working in the field. In particular, E6 mentioned that their choice of EGG was due to its connection with brain studies in prior works – ‘electrogastrography ... already had fMRI studies ... with emotion’, and E5 mentioned being motivated to work on bowel sounds based on another group’s work – ‘they [Researcher’s group] are using gut sounds to discriminate IBS ... I realized that this has ... a clinical application’. Beyond sensing, two experts discussed the interpretability of these signals to end users. E4 mentioned that ‘people were also tinkering with their own body ... trying to alter the kind of sounds that the body generated’ and that ‘how you perceive the sounds in your gut also changes culturally’. With regards to EGG signals, E6 mentioned that ‘users do not understand the signal implicitly’.

In summary, the interviews reveal limitations in existing systems’ hardware, software, and experimental capabilities, which inform the design of our OPENGUT platform.

4 OPENGUT Platform

Our literature review and formative study informed design requirements addressing key challenges in gut research workflows (see Table 2, Appendix A). The OPENGUT platform is designed to address these requirements through three integrated components.

4.1 OPENGUT Hardware

Rationale: We build a custom board that allows researchers full control over the recorded gut signals (Theme 3) while also providing a wearable belt that can hold all the electronics (Theme 1). For the

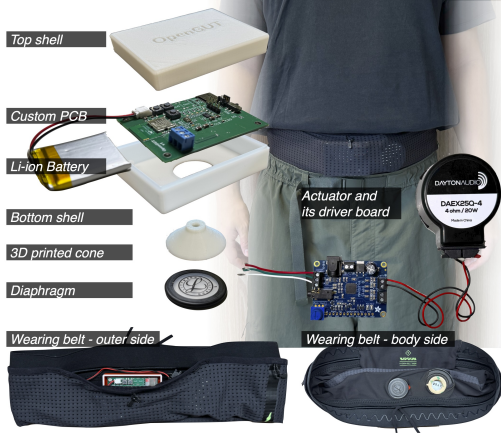


Figure 2: Exploded view of OPENGUT hardware: custom PCB, battery, 3D-printed case, haptic actuator, and belt assembly.

first iteration of our platform, we focus on capturing and rendering *bowel sounds*, rather than EGG signals. This design choice is informed by *Theme 4* which highlights the interpretability of gut sounds and their potential to support interoception research.

OPENGUT’s hardware consists of two parts: (a) embedded system, that includes a custom printed circuit board (PCB) and associated firmware to support gut sensing and actuation, and (b) body attachments, which include a 3D-printed mic-to-body interface for impedance matching as well as a stretchable belt wearable.

4.1.1 Embedded System. The custom PCB integrates a Nordic system on chip (SoC), a LiPo battery, two microphones, a micro-SD card slot for local storage, and plug-and-play actuators (Figure 2).

Custom PCB. Given its low-power consumption and extensive development support, we choose the nRF52840 chip by Nordic Semiconductors [10, 34]. Our PCB includes two digital microphones [9]: *body-facing* and *ambient-facing* microphones, to capture body sounds and reference environmental noise, respectively. The synchronized, dual-microphone configuration supports downstream noise cancellation [20, 47]. The microphones also have a frequency range of 100 Hz to 10 kHz, making them suitable for sensing bowel sounds. **Plug-and-Play Actuators.** Prior work on gut interoception have incorporated audio [33, 36] as well as haptic [26, 27] feedback methods. To support both options, we interface our PCB with audio as well as haptic actuators [11]. We choose a *plug-and-play* interfacing to keep the overall wearable lightweight, especially for long-duration audio recording (i.e., without any actuation). In addition to playback, we also support real-time visualization by connecting the audio output to a computer (via free software such as *Audacity* [4]).

4.1.2 Body Attachments. This includes two components:

Mic-to-Body Interface. As microphones are optimized for airborne vibrations, directly capturing body sounds introduces an impedance mismatch [29]. We address this by utilizing a medical-grade stethoscope diaphragm [22] to improve acoustic coupling. To bridge the dimensionality mismatch between the diaphragm and microphone inlet, we 3D-print a conical structure that enables gradual acoustic transition and improves impedance matching.

Wearable Belt. We secure the hardware with a stretchable runner’s belt, similar to Li et al. [20], featuring compartments that house the PCB, haptic actuator, amplifier, and battery. [This belt is available in multiple sizes, and is intended to provide a snug fit.](#)

4.2 OPENGUT Software Interface

Rationale. We provide a graphical user interface (GUI) that supports audio visualization and noise suppression (*Theme 2*). We also support configuration tuning for recording/playback. We utilize Python’s qt library for GUI design.

4.2.1 User Interface Features. This includes three components.

Audio Visualization and Annotation. The interface provides time series and spectrogram visualizations of recorded signals. Users can inspect recordings, segment relevant portions, annotate data, and save for further analysis. This feature set supports exploratory workflows and qualitative inspection of gut signals.

Noise Suppression. The interface provides support for high-pass filtering of body sounds (e.g., heartbeat) [5], and denoising of background noise (e.g., conversations). We integrate a pretrained state-of-the-art source separation model to extract gastric sounds [23], with options to visualize and listen to processed and raw signals.

Configuration Tuning. The interface allows users to configure recording and playback parameters, including sampling rate, recording duration, and playback intervals. These settings are written to a config file, on the micro SD-card, which is read by the firmware to enable rapid iteration and experimentation.

4.3 Documentation and Usage Modes

Rationale. Based on the gap identified from literature (Table 1), our platform is fully open-sourced and available on GitHub².

Documentation. We provide hardware design artifacts (PCB, firmware, mechanical designs) as well as software code (audio processing and GUI interface). We include quick start guide for different usage modes, which we elaborate below.

Usage Modes. OPENGUT supports three modes:

- (i) **Logger mode** enables long-term gut sound recording. Researchers can configure recording parameters through the user interface and use the PCB for continuous audio capture. Data is stored locally for subsequent analysis, including visualization and noise suppression.
- (ii) **Playback mode** allows researchers to deliver previously recorded gut signals back to users through audio or haptic channels. Playback parameters are configured via the user interface, with audio files loaded onto the PCB’s local storage.
- (iii) **Real-time Loopback Mode** enables live sensing and rendering of gut signals. Microphone input is directly transmitted to haptic actuators or headphones, without requiring access to storage.

5 Technical Evaluation

We evaluate OPENGUT along three key dimensions: signal quality, long-term recording, and battery performance in continuous use.

5.1 Signal Quality Assessment

We assessed gut-sound fidelity by comparing OPENGUT and Littmann Core stethoscope [21] recordings under controlled conditions. Since

²<https://github.com/OpenGUT/OpenGUT/>

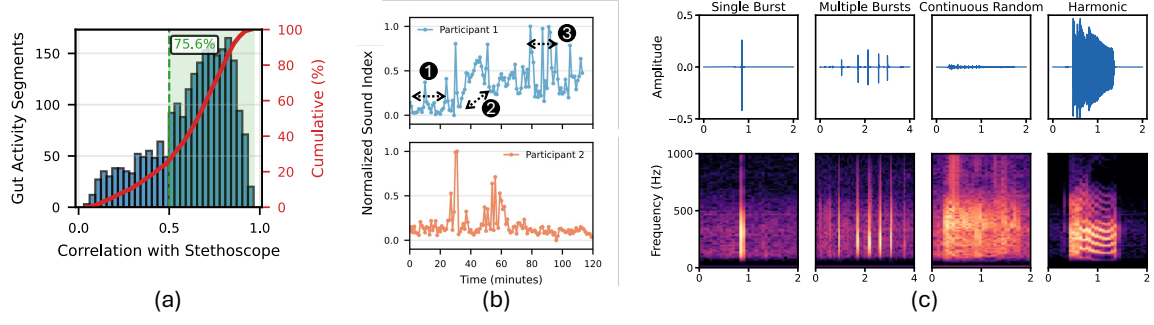


Figure 3: Technical evaluation of OPENGUT. (a) correlation between OPENGUT and stethoscope for gut activity segments, (b) gut activity trends for two-hour recordings across two participants ((1) low activity, (2) increasing activity, and (3) high activity), (c) four types of bowel sounds (single burst, multiple bursts, continuous random and harmonic sounds) from Participant 2’s data.

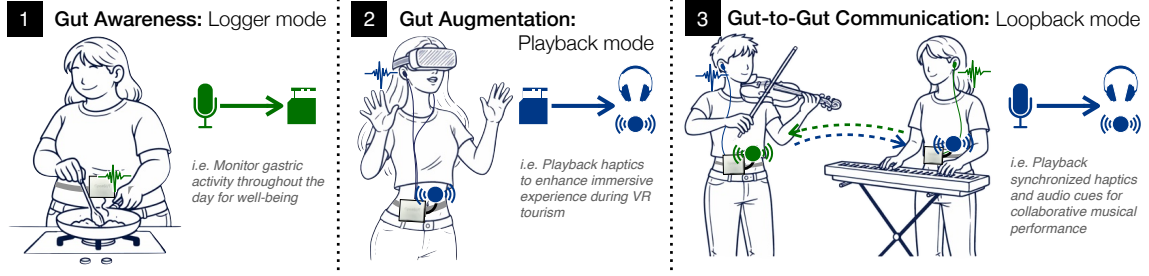


Figure 4: Novel interaction possibilities enabled by OPENGUT’s operating modes: gut awareness through logging, gut augmentation through playback, and gut-to-gut communication through loopback.

the stethoscope does not give access to raw signals, we attach an Audio Technica ATR3350xiS lavalier microphone [38], with a frequency response of 50 Hz to 18 kHz, to the stethoscope’s front-end to capture audio data. We perform recordings in an anechoic chamber to minimize the influence of external noise.

To analyze signal quality, we collected data from four participants (two female), with an average age of 32.3 years ($\sigma = 14.1$ years), while being seated in the anechoic chamber. On average, we record 20.3 minutes ($\sigma = 0.95$ minutes) each. Each participant wore both the belt that holds OPENGUT’s PCB as well as the stethoscope in the lower right quadrant of the abdomen which is the most active region for bowel sounds [13]. To compare recordings, we applied a band-pass filter from 100 Hz to 1 kHz, and identified audio segments, in multiples of 100 ms, that have significant gut activity in either the stethoscope or OPENGUT recording. We determine gut activity based on a signal threshold (empirically fixed to an amplitude of 0.005). Subsequently, we computed time-series based correlation (highest correlation within a lag window of ± 3.2 ms) between the audio segments, from stethoscope and OPENGUT microphone, respectively. Figure 3(a) shows the correlation scores across 2343 segments across all users. From the figure, we observe that 75.6% of segments have correlation of 0.5 or higher, depicting the efficacy of our platform in capturing bowel sounds. **Despite individual variability, at least 70% of the gut-activity segments from each participant had correlation scores of 0.5 or higher.** See Appendix B for per-participant results and examples of low/high correlation.

5.2 Longitudinal Gut Activity Monitoring

To demonstrate OPENGUT’s ability for longitudinal gut activity monitoring (*Theme 1*), we captured two-hour recordings in home and office settings, from two participants (one female), with a mean age of 26 years ($\sigma = 4.24$). We recorded the data in a fasting state to observe gut motility variations as suggested in the literature [12, 40]. Based on the ambient-microphone from OPENGUT, we remove noisy segments (70 segments in total with a mean duration of 7.1 seconds), apply band-pass filter, and plot the gut activity over time based on the normalized sound index (NSI) metric, proposed by Tomomasa et al. [40]. We computed sound index as the sum of absolute signal amplitudes within a 1-minute time-segment, and performed min-max normalization over all segments, to obtain the *normalised sound index*, which we plot in Figure 3(b). As shown, the gut activity captured by our platform demonstrates various phases in gut activity: low activity, increasing activity, and high activity. Figure 3(c) shows the four well-known types of bowel sounds captured by our microphone. **This experiment provides initial evidence of OPENGUT’s feasibility for longitudinal gut sensing,** and for designing interventions based on gut activity levels as well as bowel-sound types.

To assess the suitability of our hardware for long-term monitoring, we performed battery estimation tests. Specifically, we used Nordic’s Power Profiler kit II [28], to profile OPENGUT, for a period of one hour each in the three modes, to estimate the battery life.

Based on the results, we can support 24 hours each of logging, playback and loopback, with battery capacities of 1000 mAh, 1720 mAh and 1320 mAh, respectively, while accounting for a 30% tolerance. Based on the application requirements, researchers can choose an appropriate trade-off between battery capacity and its weight.

6 Limitations and Future Work

Limitations. OPENGUT provides an open-source toolkit that enables HCI researchers to explore gut-centered interactions. While our work establishes the feasibility of such a platform, we acknowledge that further validation is needed. Although our technical evaluation was conducted in a controlled low-noise environment, OPENGUT is designed with future real-world deployment in mind with the inclusion of a dedicated ambient-facing reference microphone and an integrated denoising pipeline, providing a good starting point for addressing environmental noise and motion artifacts.

We have also taken several steps toward reproducibility by open-sourcing the hardware, firmware, software, and providing detailed documentation through the OPENGUT website. While the current paper does not include a formal usability or reproducibility study, we view this release as the first step toward a community-driven platform. As part of the next iteration, we plan to recruit researchers to provide feedback on how to improve the OPENGUT platform and facilitate validation in different use scenarios.

The OPENGUT toolkit also currently lacks Bluetooth data transfer and semi-automated annotation, features that could substantially reduce researchers' workload. We plan to incorporate these capabilities in future iterations.

Future Work. We envision three interaction classes with OPENGUT: (i) Gut Awareness, (ii) Gut Augmentation, and (iii) Gut-to-Gut Communication, supported by the usage modes of OPENGUT, namely, logger, playback and real-time loopback, respectively, described in Section 4.3. Given that the interaction classes need to be validated, we consider them as part of future work. (i) Gut Awareness: Sense gut sounds over long periods of time, enabling users to self-reflect on the relationship between their food intake or emotional states, with their gut activity, and elevate their bodily awareness [1] (Figure 4(a)). (ii) Gut Augmentation: Amplify existing gut sensations to improve interoception, or enable new gut sensations for creating novel experiences. For instance, in immersive movie watching [2], similar to an audio track, we envision a 'visceral track', that encodes haptic signals targeting the gut, for eliciting emotional experiences [27], in synchrony with the movie content (Figure 4(b)). (iii) Gut-to-Gut Communication: Sense gut activity from one user and provide real-time feedback to another user, creating a new embodied communication channel, e.g., for collaborative art. In live music improvisation [30, 37], we can envision sensing one musician's gut activity and rendering it as haptic feedback to another, conveying internal physiological states that align with musical expression (Figure 4(c)).

Overall, our work takes a step toward realizing an open gut platform, empowering our community to advance gut research.

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A Design Requirements

Table 2 presents the three design requirements derived from the formative study as well as the literature, covering different aspects within the hardware, software as well as the documentation components of OPENGUT platform.

1. Research-Oriented Wearable Hardware (based on Themes 1,3 and 4)	
1a. Custom PCB with microphones	Design PCB with dual microphone (gut-facing and ambient-facing) and support for local storage of raw signals with SD card
1b. Plug-and-play actuators	Connect PCB with external amplifier board and actuator-of-choice for intervention (haptic transducer, audio headphones)
1c. Extensible firmware	Create firmware that supports various modes of operation, including logging, playback and real-time loopback
1d. 3D-printed acoustic frontend	Minimize impedance mismatch between body and microphone using 3D-printed conical structure and stethoscope diaphragm
1e. Belt design	Design belt to secure PCB in place and microphone in contact with the body
2. Interactive Software for Signal Exploration and Analysis (based on Theme 2)	
2a. Updating configurations	Enable updating recording (duration, sampling rate) and playback settings (duration)
2b. Audio visualization and annotation	Plot time-series and spectrogram of recordings, annotate segments
2c. Noise cancellation support	Apply high-pass filters and denoising machine learning models to recordings
3. Open Documentation and Implementation Resources (based on literature presented in Table 1)	
3a. Quick start guide	Provide template code for recording and playback
3b. Usage mode-wise documentation	Provide detailed description of each mode of operation, and options for customization

Table 2: Design requirements derived from the literature review and formative-study themes.

B Technical Evaluation: Extended Details

As shown in Figure 5, at least 70% of gut activity segments from each participant have high correlation (≥ 0.5), highlighting OPENGUT's audio capture efficacy. We also observe that the total number of gut activity segments differs across participants: 605, 989, 123 and 626, across participants A-D, respectively. The number of segments identified are both dependent on the individual's gut activity as well as the chosen threshold (empirically fixed to be 0.005) to detect gut activity. One limitation of our analysis is the reliance on automatic segmentation of gut activity, which is less accurate than manual annotation. However, given that our experiments were conducted in a low-noise environment (i.e., an anechoic chamber), this approach remains reasonable for our setting.

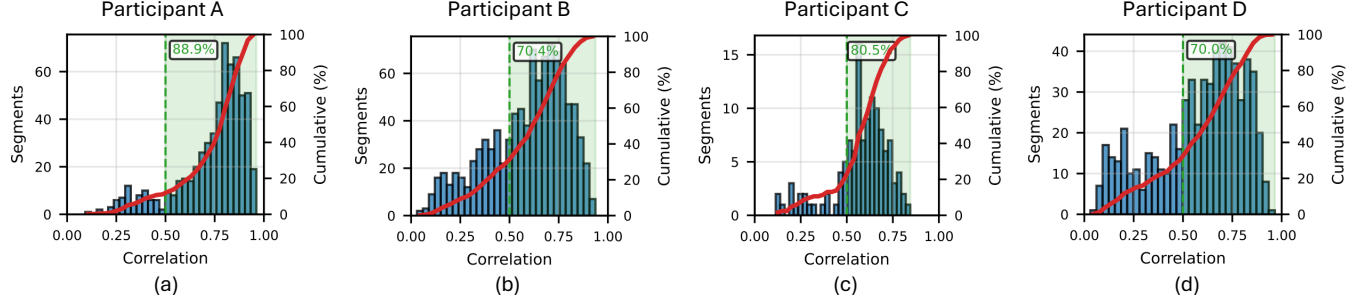


Figure 5: Correlation scores for gut activity segments between OPENGUT and stethoscope recordings for each of the four participants, part of the signal quality assessment experiment.

Figure 6 depicts two example audio segments from Participant D's data, with high and low correlation. In the high correlation case (correlation score = 0.93), the time-series trends are similar despite the differences in signal amplitude. In contrast, the low correlation instance (correlation score = 0.06) reflects noise captured by OPENGUT microphone (e.g., breathing or shirt movement), highlighting the need for noise cancellation.

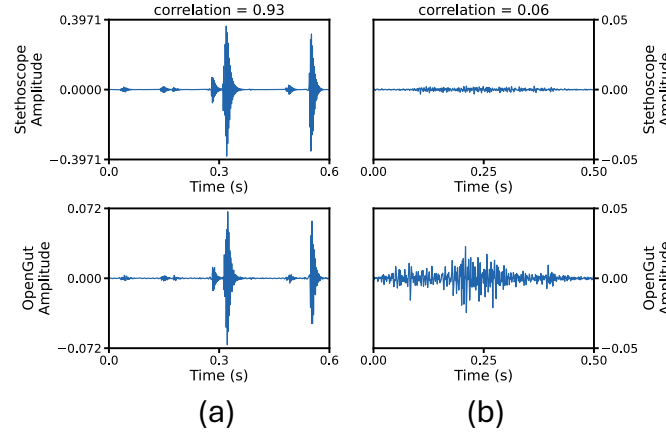


Figure 6: Examples of (a) high correlation (0.93) and (b) low correlation (0.06) between stethoscope and OPENGUT recordings.